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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
The experiment was repeated three times.
Number: The experiment was repeated three times.
Cell Line Authentication

not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Materials and Methods Materials Chitosan (Ch, low molecular weight, catalogue no. 448869, Sigma-Aldrich, Pozna#, Poland), glycidyltrimethylammonium chloride (GTMAC, ≥90%, catalogue no. 50053, Sigma-Aldrich, Warsaw, Poland), iron (II) chloride tetrahydrate (puriss. p.a., ≥98.5%, catalogue no. 44939, Sigma-Aldrich, Pozna#, Poland), iron (III) chloride hexahydrate (puriss. p.a., ≥98.5%, catalogue no. 31232-M, Sigma-Aldrich, Pozna#, Poland), p-toluenesulfonyl chloride (tosyl chloride, TsCl, ReagentPlus®, ≥99%, catalogue no. 240877, Sigma-Aldrich, Pozna#, Poland), 1,10-phenanthroline monohydrate (p.a., catalogue no. 413000221, POCH S.A.), polyethylenimine branched (PEI, average Mw ~25,000 by LS, Sigma-Aldrich, Pozna#, catalogue no. 408727), t-octylphenoxypolyethoxyethanol (TritonTM X-100, for molecular biology, catalogue no. T8787, Sigma-Aldrich, Pozna#, Poland), phosphate-buffered saline (PBS, tablet, catalogue no. P4417, Sigma-Aldrich, Pozna#, Poland), ethylenediaminetetraacetic acid (EDTA, ACS reagent, 99.4–100.6%, powder, catalogue no. E9884, Sigma-Aldrich, Pozna#, Poland), bovine serum albumin (BSA, BioReagent, suitable for cell culture, catalogue no. A9418, Sigma-Aldrich, Pozna#, Poland), fetal bovine serum (FBS, qualified, heat inactivated, catalogue no. 10500064, Gibco, Thermo Fisher Scientific, Warsaw, Poland), (hydroxymethyl)aminomethane (Tris; Sigma Aldrich, Pozna#, catalogue no. GE17-1321-01), sodium dodecyl sulfate (SDS; Sigma Aldrich, Pozna# catalogue no. GE17-1313-01), β-mercaptoethanol (Sigma Aldrich, Pozna#, catalogue no. 444203),	donkey IgG		
	Anti-Sheep IgG		
	Anti-E-cadherin		
	β-actin		
	Anti-rabbit IgG		

glycerol (for molecular biology, ≥99.0%, Sigma Aldrich, Pozna#, catalogue no. G5516), Tween® 20 (viscous liquid, Sigma Aldrich, Pozna#, catalogue no. P1379), fluorescein isothiocyanate (FITC, BioReagent, ≥90%, catalogue no. 46950, Sigma-Aldrich, Pozna#, Poland), Human/Mouse/Rat N-Cadherin Antibody (polyclonal sheep IgG, catalogue no. AF6426, R&D Systems), Donkey Anti-Sheep IgG NorthernLights™ NL557-conjugated Antibody (polyclonal donkey IgG, catalogue no. NL010, R&D Systems, Bio-Techne, Warsaw, Poland), Anti-Sheep IgG HRP-conjugated (R&D Systems, catalogue no. XDP 1421031), Anti-E-cadherin (Cell Signaling Technology, catalogue no. 3195S), β-actin (Cell Signaling Technologies, catalogue no. 4970S), Anti-rabbit IgG HRP-linked (Cell Signaling Technology, catalogue no. 7074), PC-3 cell line (ATCC®, catalogue no. CRL-1435, LGC Standards, #omianki, Poland), DU 145 cell line (ATCC®, catalogue no. HTB-81, LGC Standards, #omianki, Poland), LNCaP clone FGC cell line (ATCC®, catalogue no. CRL-1740, LGC Standards, #omianki, Poland), RPMI 1640 culture medium (Gibco, catalogue no. 21875034, ThermoFisher Scientific, Warsaw, Poland), Qubit Protein Assay Kit (Thermo Fisher Scientific, catalogue no. Q33211), 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT, powder, BioReagent, ≥97.5% (HPLC), catalogue no. M5655, Sigma-Aldrich, Pozna#, Poland), Resazurin sodium salt (Alamar Blue, powder, BioReagent, catalogue no. R7017, Sigma-Aldrich, Pozna#, Poland), 4#,6-diamidino-2-phenylindole (DAPI, for nucleic acid staining, catalogue no. D9542, Sigma-Aldrich, Pozna#, Poland), were all used as received.			
Samples of PC-3 cells, DU 145 cells and LNCaP cells were prepared and incubated with fluorescently labeled SPION/CCh/N-cad (the number of	anti-N-cadherin		

added nanoparticles corresponded to the anti-N-cadherin antibody concentration of 2.5 µg/mL) for 1 hour at 4 °C.			
Experimental Models: Cell Lines			
The obtained results have confirmed that the level of N-cadherin is the highest for the PC-3 cells, lower for DU 145 and no N-cadherin was observed for LNCaP cell line.	PC-3		Suggestion: RRID:CVCL_0035(link)
Co-cultures of cells: PC-3 and DU 145 (cell number ratio 1:1), PC-3 and LNCaP (cell number ratio 1:4) were obtained by simultaneous seeding of cell mixtures on glass coverslips and incubated until a monolayer of cells was observed.	LNCaP		Suggestion: RRID:CVCL_0395(link)
Samples of PC-3 cells, DU 145 cells and LNCaP cells were prepared and incubated with fluorescently labeled SPION/CCh/N-cad (the number of added nanoparticles corresponded to the anti-N-cadherin antibody concentration of 2.5 µg/mL) for 1 hour at 4 °C.	DU 145		
Software and Algorithms			
Materials and Methods Materials Chitosan (Ch, low molecular weight, catalogue no. 448869, Sigma-Aldrich, Pozna#, Poland), glycidyltrimethylammonium chloride (GTMAC, ≥90%, catalogue no. 50053, Sigma-Aldrich, Warsaw, Poland), iron (II) chloride tetrahydrate (puriss. p.a., ≥98.5%, catalogue no. 44939, Sigma-Aldrich, Pozna#, Poland), iron (III) chloride hexahydrate (puriss. p.a., ≥98.5%, catalogue no. 31232-M, Sigma-Aldrich, Pozna#, Poland), p-toluenesulfonyl chloride (tosyl chloride, TsCl, ReagentPlus®, ≥99%, catalogue no. 240877, Sigma-Aldrich, Pozna#, Poland), 1,10-phenanthroline monohydrate (p.a., catalogue no. 413000221, POCH S.A.), polyethylenimine branched (PEI, average Mw ~25,000 by LS, Sigma-Aldrich, Pozna#, catalogue no. 408727), t-octylphenoxypolyethoxyethanol (TritonTM X-100, for molecular biology, catalogue no. T8787,	Tris; Sigma Aldrich		
	SDS; Sigma Aldrich		
	Gibco		

Sigma-Aldrich, Pozna#, Poland), phosphate-buffered saline (PBS, tablet, catalogue no. P4417, Sigma-Aldrich, Pozna#, Poland), ethylenediaminetetraacetic acid (EDTA, ACS reagent, 99.4–100.6%, powder, catalogue no. E9884, Sigma-Aldrich, Pozna#, Poland), bovine serum albumin (BSA, BioReagent, suitable for cell culture, catalogue no. A9418, Sigma-Aldrich, Pozna#, Poland), fetal bovine serum (FBS, qualified, heat inactivated, catalogue no. 10500064, Gibco, Thermo Fisher Scientific, Warsaw, Poland), (hydroxymethyl)aminomethane (Tris; Sigma Aldrich, Pozna#, catalogue no. GE17-1321-01), sodium dodecyl sulfate (SDS; Sigma Aldrich, Pozna# catalogue no. GE17-1313-01), β-mercaptoethanol (Sigma Aldrich, Pozna#, catalogue no. 444203), glycerol (for molecular biology, ≥99.0%, Sigma Aldrich, Pozna#, catalogue no. G5516), Tween® 20 (viscous liquid, Sigma Aldrich, Pozna#, catalogue no. P1379), fluorescein isothiocyanate (FITC, BioReagent, ≥90%, catalogue no. 46950, Sigma-Aldrich, Pozna#, Poland), Human/Mouse/Rat N-Cadherin Antibody (polyclonal sheep IgG, catalogue no. AF6426, R&D Systems), Donkey Anti-Sheep IgG NorthernLights™ NL557-conjugated Antibody (polyclonal donkey IgG, catalogue no. NL010, R&D Systems, Bio-Techne, Warsaw, Poland), Anti-Sheep IgG HRP-conjugated (R&D Systems, catalogue no. XDP 1421031), Anti-E-cadherin (Cell Signaling Technology, catalogue no. 3195S), β-actin (Cell Signaling Technologies, catalogue no. 4970S), Anti-rabbit IgG HRP-linked (Cell Signaling Technology, catalogue no. 7074), PC-3 cell line (ATCC®, catalogue no. CRL-1435, LGC Standards, #omianki, Poland), DU 145 cell line (ATCC®, catalogue no. HTB-81, LGC Standards, #omianki, Poland), LNCaP clone FGC cell line (ATCC®, catalogue no. CRL-1740, LGC Standards, #omianki, Poland), RPMI 1640				

culture medium (Gibco, catalogue no. 21875034, ThermoFisher Scientific, Warsaw, Poland), Qubit Protein Assay Kit (Thermo Fisher Scientific, catalogue no. Q33211), 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT, powder, BioReagent, ≥97.5% (HPLC), catalogue no. M5655, Sigma-Aldrich, Pozna#, Poland), Resazurin sodium salt (Alamar Blue, powder, BioReagent, catalogue no. R7017, Sigma-Aldrich, Pozna#, Poland), 4#,6-diamidino-2-phenylindole (DAPI, for nucleic acid staining, catalogue no. D9542, Sigma-Aldrich, Pozna#, Poland), were all used as received.			
The images were collected and analyzed using NIS-Elements confocal software.	NIS-Elements		
The data obtained were recorded using BDFACSDivaTM v9.0 and analyzed using FlowJoTM v10.8 program to determine the percentage and mean fluorescence intensity of positive cells.	BDFACSDivaTM		
	FlowJoTM		

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

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